

Mid Semester Examination**Digital Circuits and Systems**Duration: 1 Hour 30 min.
Marks: 15

/ INECC01

Note: Attempt all five questions. Missing data/information (if any) may be suitably assumed and mentioned in the answer.

1.	(a)	Given that $292_{10} = 1204_b$. Find the value of b.	1.5 Marks	CO1
	(b)	Design the equivalent expression of the following using minimum number of universal gates. (i) $\bar{A} \oplus \bar{B}$ (ii) $\bar{A} \odot \bar{B}$	1.5 Marks	CO1
2.	(a)	Design a BCD to excess-3 code converter.	1.5 Marks	CO2
	(b)	Implement the following function using 8:1 multiplexer $F(A, B, C, D) = \sum m(1, 2, 5, 6, 8, 9, 13, 14, 15)$ Assume B, C, and D as the selection variables.	1.5 Marks	CO2
3.	(a)	Implement a full adder using 3×8 decoder.	1.5 Marks	CO2
	(b)	Minimize the following expression using K-Map and realize with gates $F(A, B, C, D) = \sum m(1, 2, 3, 5, 6, 7, 9, 10, 11, 13, 14, 15)$	1.5 Marks	CO1
4.	(a)	Design and implement a 2-bit comparator for the binary numbers A (A_1A_0) and B (B_1B_0) using suitable logic gates.	1.5 Marks	CO1
	(b)	Implement a 4×16 decoder using 2×4 decoders.	1.5 Marks	CO2
5.	(a)	Explain the operational characteristics of BJT as a switch.	1.5 Marks	CO3
	(b)	Define the characteristics of logic families.	1.5 Marks	CO3